often administered at a location distant from its intended site of action. Therefore, a useful drug must have necessary properties to be transported from its site of administration to its site of action.

The Physical Nature of a Drug

Drugs may be solid at room temperature (e.g, aspirin, atropine), liquid (e.g, nicotine, ethanol), or gaseous (e.g, nitrous oxide). These factors often determine the best route of administration.

Drug Size

The molecular size of the drug varies from very small to larger one. If the molecular size of the drug is very small it easily diffuses the cellular compartments, in case of larger, the drug is administered directly into the vascular compartment by intravenous or intra-arterial infusion.

Drug reactivity and Drug-receptor bonds

Drugs interact with receptors by means of chemical forces or bonds. There are three major types like Covalent, Electrostatic, and Hydrophobic.

Drug Shape

The Shape of a drug molecule must be such as to permit binding to its receptor site via the bonds.

9.4 SOURCES OF DRUGS

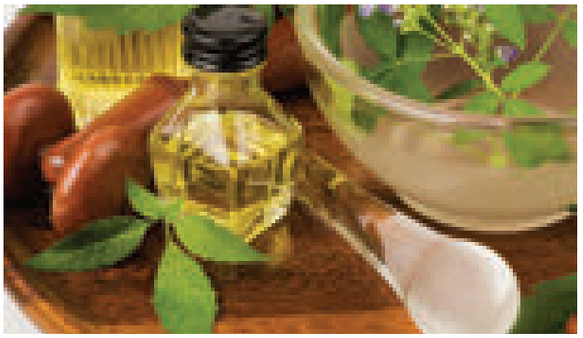
Drugs are obtained from various sources like plants, animals, microbial, mineral, synthetic and semi synthetic after repeated extraction and purification.

Plant Sources

Most of the drugs are obtained from the plant sources.

E.g. Morphine extracted from the plant OPIUM through the principle of ALKALOID.

Opium Alkaloid Extraction



Animal Sources

Microbial Source: *Example:* **Penicillin**

Mineral Source: *Example:* **Aluminium Hydroxide**.

Synthetic Source: *Example:* **Analgesics**

Semi Synthetic Source: *Example:* **Atropine**

Animals

Animal	Part	Drug	Use
Cow	Pancreas	Insulin	Antidiabetic hormaone
Fish	Sperms	Protamine sulphate	Antidote of heparin
Pig	Intestine	Heparin	Anticoagulants
Ox	Lungs	Heparin	Anticoagulants



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